DEPARTMENT OF PHYSICS · **UDESC**

SEMINAR · **2026**

RAY TRACING IN RELATIVISTIC STARS

From geometry to *observation*

How spacetime curvature decides the *image*, the *spectrum* and the *light curve* that an X-ray telescope receives from a neutron star.



$$\psi(b) = \int du \,/\, \sqrt{(b^{-2} - u^2(1 - R_S u))}$$ THE LIGHT LEAVING THE SURFACE

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OUTLINE

Three parts, *one path*

I

Relativistic ray tracing

Null geodesics in Schwarzschild: light bending, redshift and time delays — and how to follow the rays backwards, from the telescope to the surface.

II

Pulsars and neutron stars

The laboratories: structure and equation of state, pulsars and magnetars, hot spots, rotation, atmospheres — and what NICER has already measured.

III

Recent results

My published work: magnetar light curves, a massive low-field magnetar, accretion columns — and tests of alternative theories of gravity.

THE PREMISE

Everything we know arrives *as photons*

Compact stars are *natural laboratories* of strong-field gravity — but we know them only through the photons that left them and crossed a curved spacetime.

metric → null geodesics → image · spectrum · light curve → telescope → **M, R, field, gravity**

THE DIRECT PROBLEM

Given the star and the metric, predict what is observed. That is **ray tracing**.

THE INVERSE PROBLEM

Given what is observed, infer mass, radius, spot geometry and magnetic field.

THE TEST

If the metric is not Schwarzschild, the light curves change: observation tests **gravity**.

I

PART ONE

Relativistic *ray tracing*

Near a mass, light does not travel in a straight line: it follows the null geodesics of the metric. Ray tracing means following those geodesics backwards — from the telescope to the source.

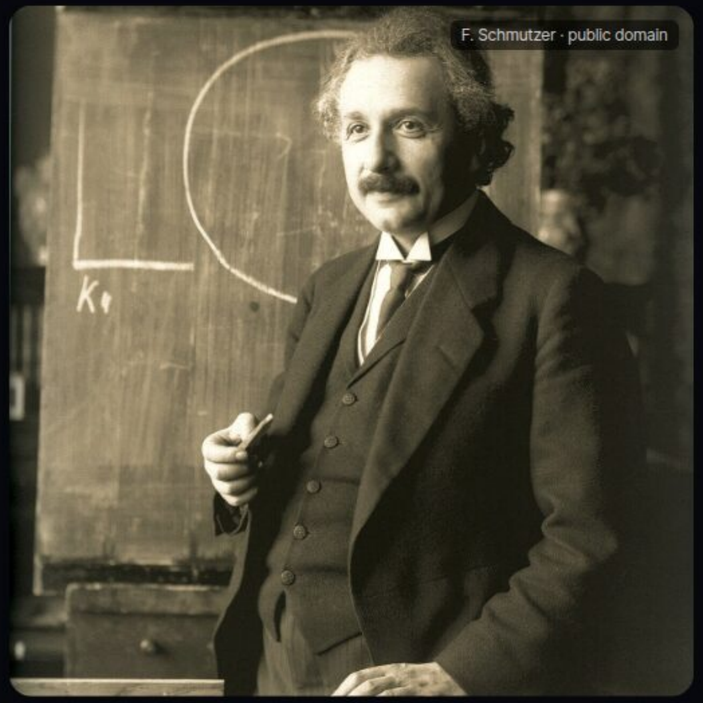
null geodesics · light bending · redshift · the image of a compact star · the recipe



SMACS 0723 (JWST) · NASA, ESA, CSA, STScI · public domain

25 NOVEMBER 1915

Einstein and *his field equations*



Albert Einstein
1921

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

CURVATURE = ENERGY AND MOMENTUM CONTENT

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0$$

GEODESIC

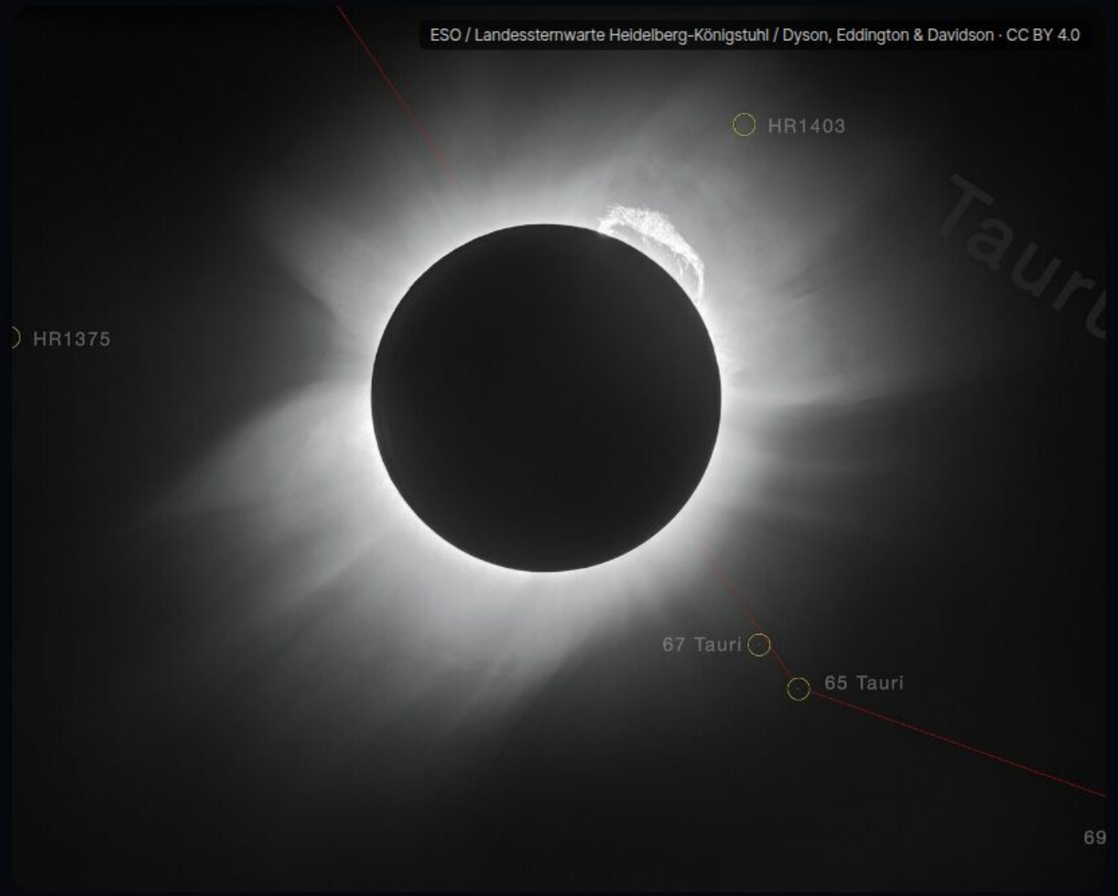
$$ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu$$

THE METRIC IS THE FIELD

> *"Spacetime tells matter how to move; matter tells spacetime how to curve."*
>
> John A. Wheeler

THE CLASSICAL TESTS

Sobral, 1919: light *bends*



EFFECT	GR PREDICTS
perihelion of Mercury (1915)	43″ / century
light grazing the Sun (1919)	1.75″
redshift, Pound–Rebka (1959)	2.5×10^{-15} over 22.5 m
Shapiro delay (1964)	~200 µs
GPS clocks	+38 µs / day

Sobral expedition (Ceará, Brazil): **1.98 ± 0.12″**. Príncipe: 1.61 ± 0.30″. The sky over north-eastern Brazil confirmed Einstein.

DECEMBER 1915 · RUSSIAN FRONT

Schwarzschild and the *first exact solution*



Karl Schwarzschild
1873 – 1916

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \frac{dr^2}{1 - r_s/r} + r^2 d\Omega^2$$

$R_s = 2GM/C^2$ · VALID OUTSIDE ANY NON-ROTATING SPHERICAL BODY

$1\ r_s$	$1.5\ r_s$	$3\ r_s$	2.95 km
event horizon	photon sphere	ISCO: last stable orbit	r_s of the Sun (Earth: 8.9 mm)

He wrote to Einstein on 22 December 1915 while serving as an artillery officer. He died in May 1916.

WHAT WE NEED FROM THE METRIC

Curvature does *three things* to a photon

1 · IT BENDS THE PATH

The photon follows a **null geodesic**. Near the surface of a neutron star the bending is tens of degrees: we see **more than half** of the star.

2 · IT CHANGES THE ENERGY

A photon emitted at r with local energy E reaches infinity with

$$E_\infty = E\sqrt{1 - R_s/r}$$

GRAVITATIONAL REDSHIFT

3 · IT DELAYS

Different paths take different times (Shapiro delay): photons from each point of the star arrive tens of microseconds apart.

Ray tracing puts all three into a single calculation, **pixel by pixel**, for each instant of the rotation.

In a neutron star, $R \approx 2.5$–$3\ R_S$: none of the three effects is a small correction.

And all three depend only on the metric outside the star — which is why a light curve carries information about gravity.

Two constants *and a potential*

The metric depends on neither t nor ϕ: energy and angular momentum are conserved.

$$E = \left(1 - \frac{R_s}{r}\right) \dot{t}$$

$$L = r^2 \dot{\varphi}$$

For light only the ratio matters, the **impact parameter** b = L/E:

$$\frac{1}{b^2} = \frac{\dot{r}^2}{L^2} + V(r), \quad V = \frac{1}{r^2}\left(1 - \frac{R_s}{r}\right)$$

A "POTENTIAL" FROM GEOMETRY ALONE

$$\left(\frac{du}{d\varphi}\right)^2 = \frac{1}{b^2} - u^2(1 - R_s u)$$

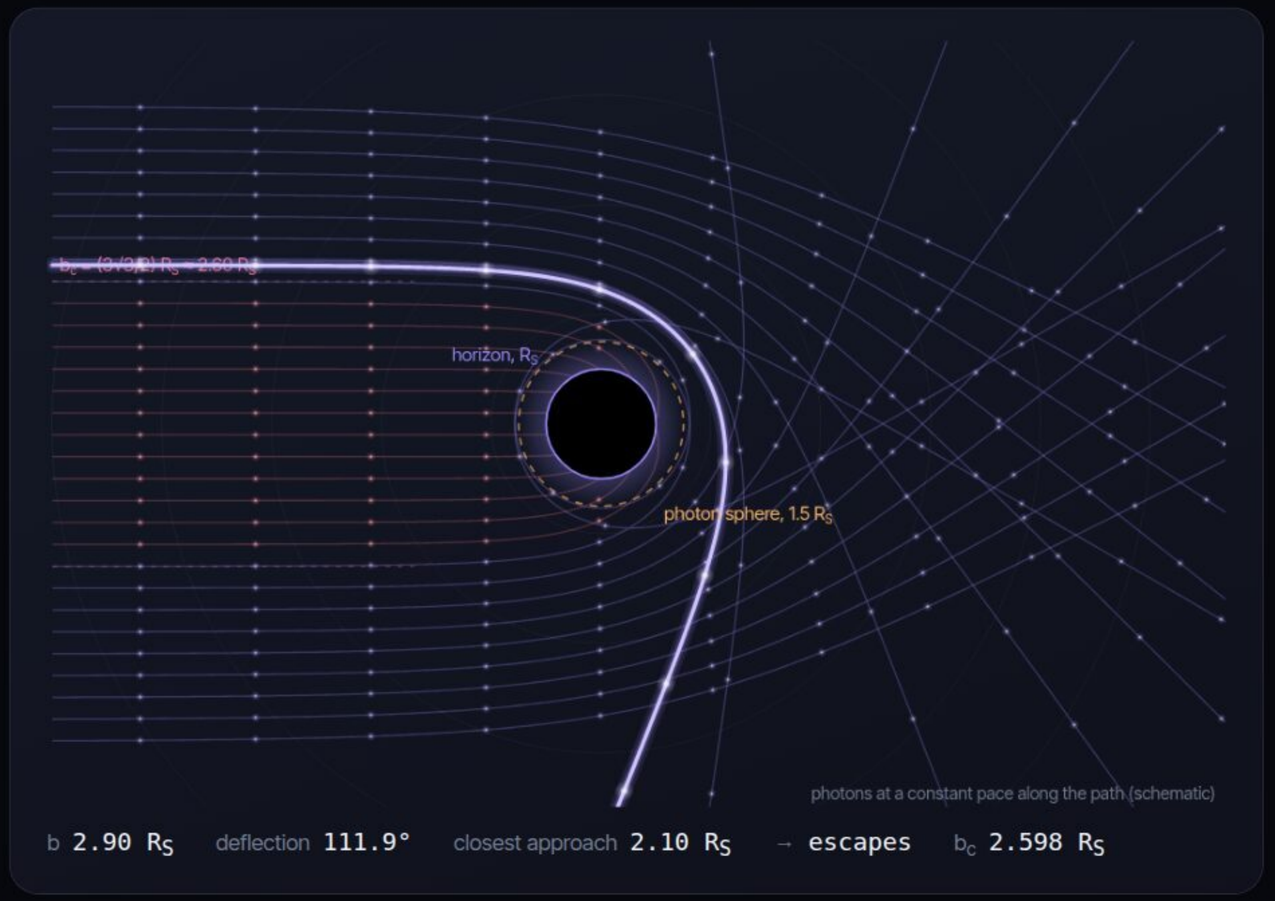
THE ORBIT · $U = 1/R$

The maximum of V lies at $r = 1.5\ R_S$: the **photon sphere**. Below $b_c = (3\sqrt{3}/2)\ R_S$ the ray comes from the horizon — or from the star.



SCHWARZSCHILD GEODESICS

How light *bends*



b 2.90 R_S deflection 111.9° closest approach 2.10 R_S → escapes b_c 2.598 R_S

$$u'' + u = \frac{3}{2} R_S u^2$$

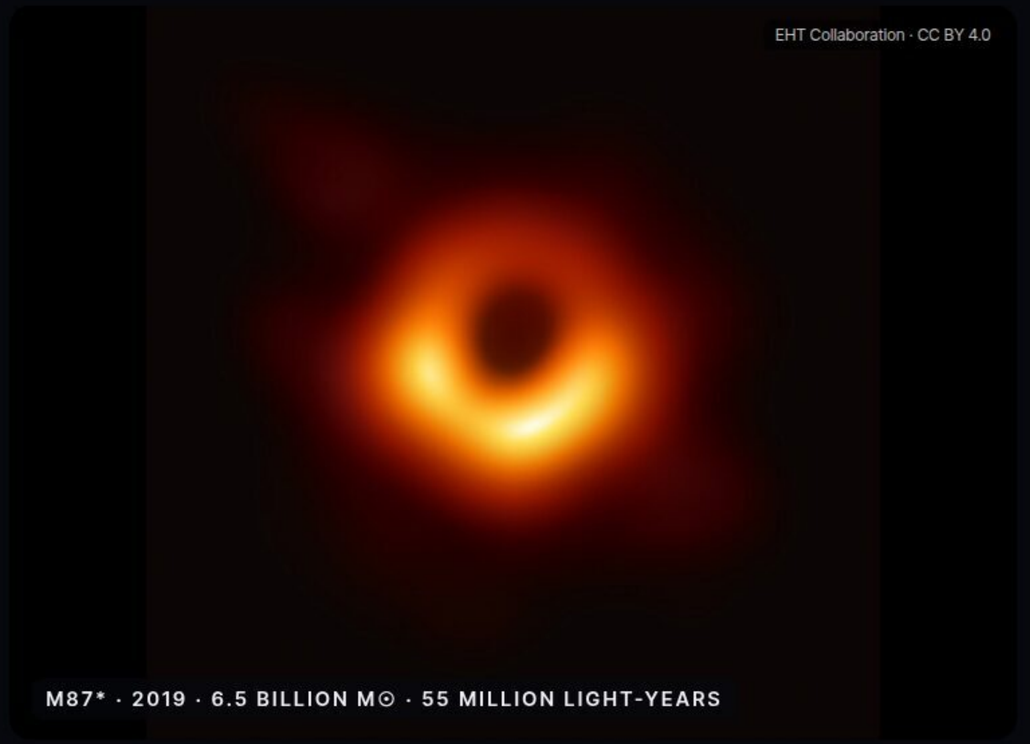
$U = 1/R$ · ORBIT EQUATION FOR LIGHT

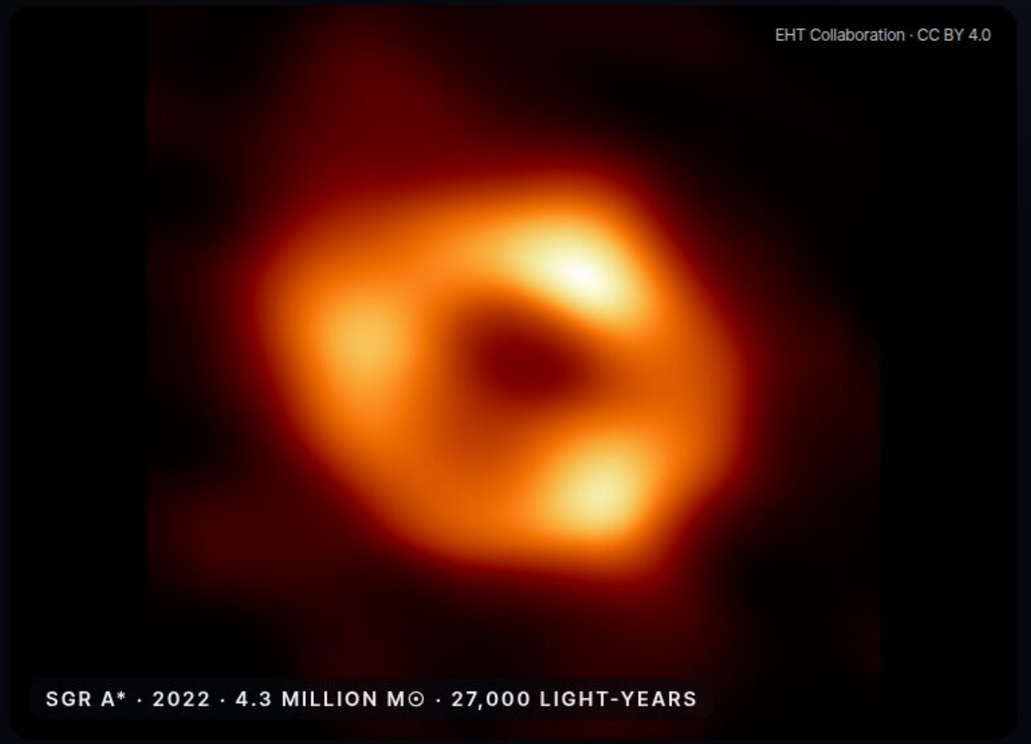
- $b > 3\sqrt{3}\,GM/c^2 \approx 2.6\ R_S$: the light is deflected and escapes.
- $b = 2.6\ R_S$: unstable orbit on the **photon sphere**, $r = 1.5\ R_S$.
- $b < 2.6\ R_S$: captured.

The term on the right is the General Relativity correction. Without it, light would travel in a straight line.

EVENT HORIZON TELESCOPE · 2019 AND 2022

The *shadows*





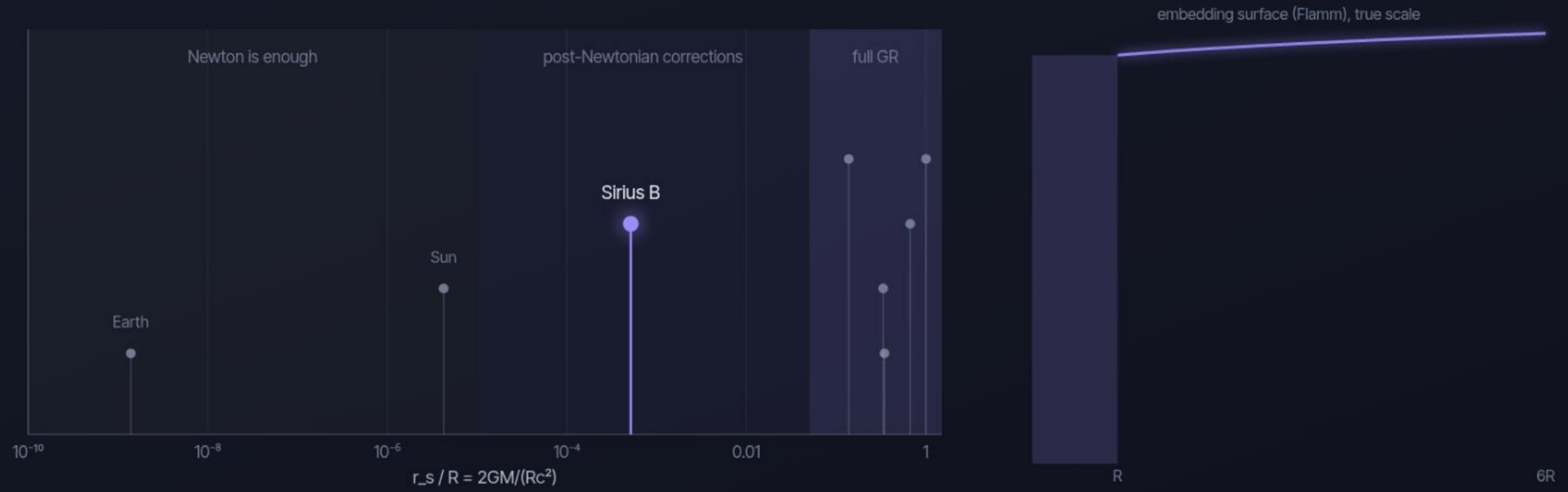
The ring is ~5 R_S across: it is the photon sphere, at 1.5 R_S, seen through the lens of the black hole itself — shadow diameter = $2\sqrt{27}\,GM/c^2 \approx 5.2\,R_S$.

COMPACTNESS

Where GR *stops being a correction*

FROM THE EARTH TO THE HORIZON · CLICK AN OBJECT

Earth | Sun | Sirius B | proto-NS | neutron star | ISCO | photon sph. | horizon



Sirius B · white dwarf, 1.02 M☉ · R = 5 840 km r_s/R `5.15e-4` redshift z `2.58e-4` clock loses `22.3` s per day grazing light deflection `213″`

$z = (1 - R_S/R)^{-1/2} - 1$ · clock at rest on the surface compared with infinity · deflection of a grazing ray from the exact orbit integral (weak field: $2R_S/R$). The objects advance on their own every few seconds.

THE ANGLE THAT MATTERS

From the surface to infinity: $\psi(\alpha)$

A photon leaves the surface at an angle α to the normal. By the time it reaches the observer, it has turned through ψ around the centre:

$$b = \frac{R \sin \alpha}{\sqrt{(1 - R_S/R)}}$$

$$\psi(b) = \int_0^{1/R} \frac{du}{\sqrt{(b^{-2} - u^2(1 - R_S u))}}$$

EXACT

$$1 - \cos \alpha \approx (1 - \cos \psi)(1 - \frac{R_S}{R})$$

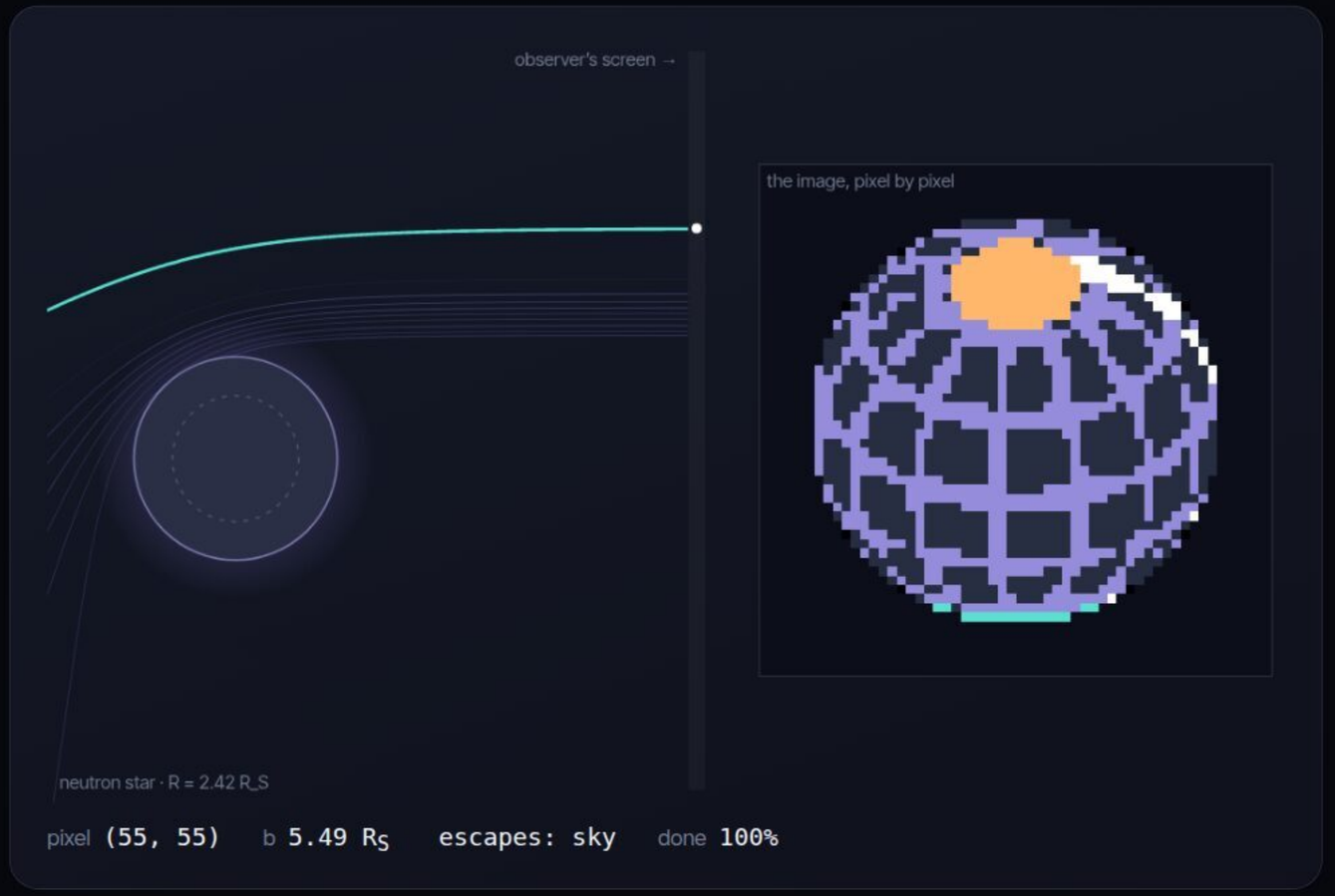
BELOBORODOV (2002)

Without gravity, ψ = α and the limb is at 90°. With gravity, the limb moves further round: **we see behind the star**. The approximation is excellent down to $R \approx 3\,R_S$ and fails precisely where the secondary image appears.



RAY TRACING BACKWARDS

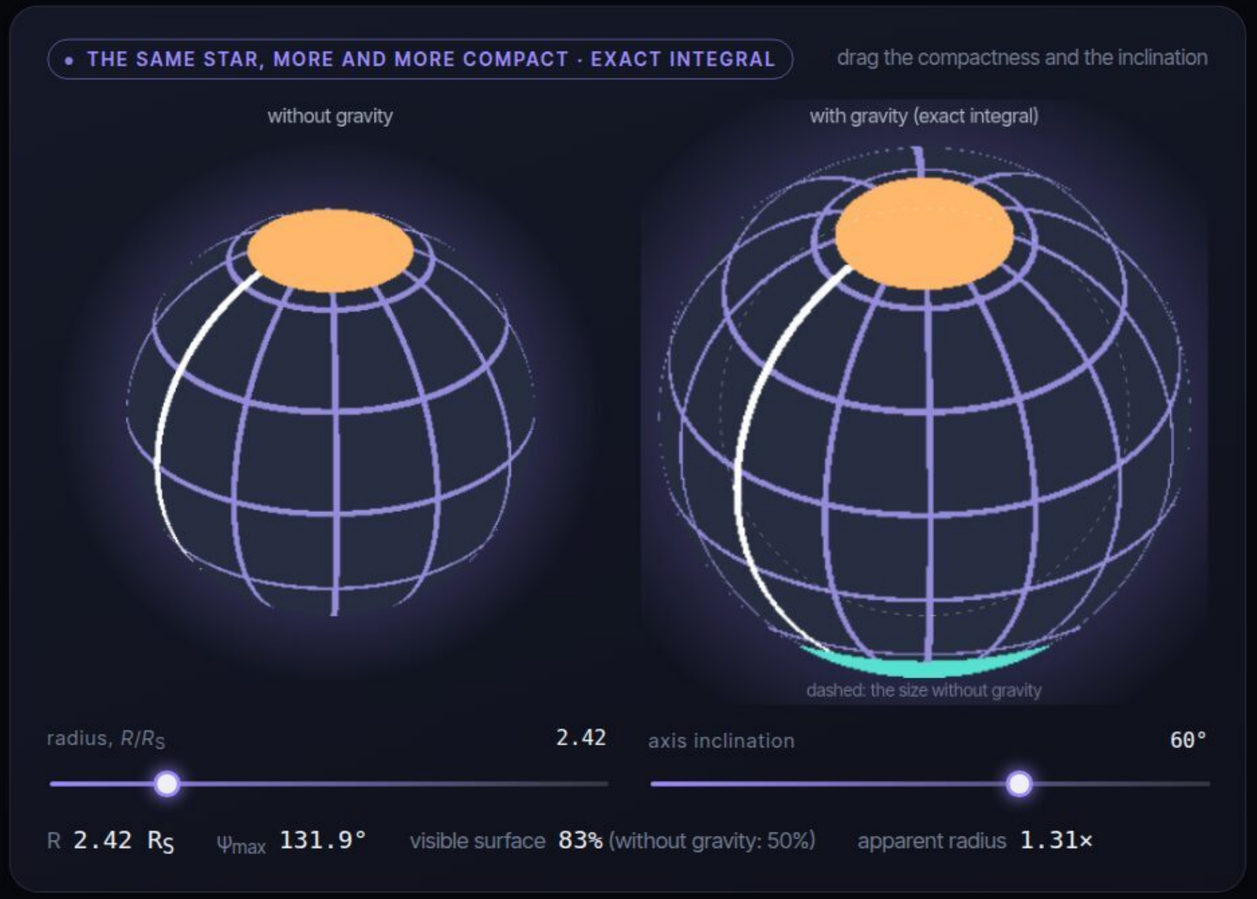
One pixel, *one ray*



- Each pixel of the image is a ray that **arrives** at the observer. Its impact parameter *b* is the distance to the centre of the image.
- The geodesic is integrated **backwards**: the ray escapes to the sky or hits the surface at some point, at an angle α.
- The pixel takes the colour of the point the light left from — corrected for redshift and for the emission angle.
- Rays that would never reach the telescope are never computed: that is what makes the method efficient.

With spherical symmetry, ψ depends only on *b*: one table ψ(*b*), computed once, serves every pixel.

We see *behind* the star

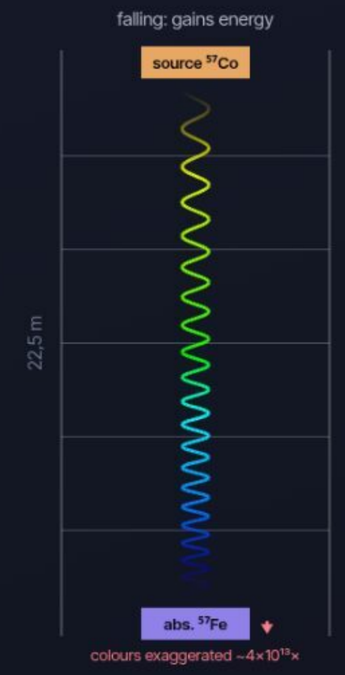


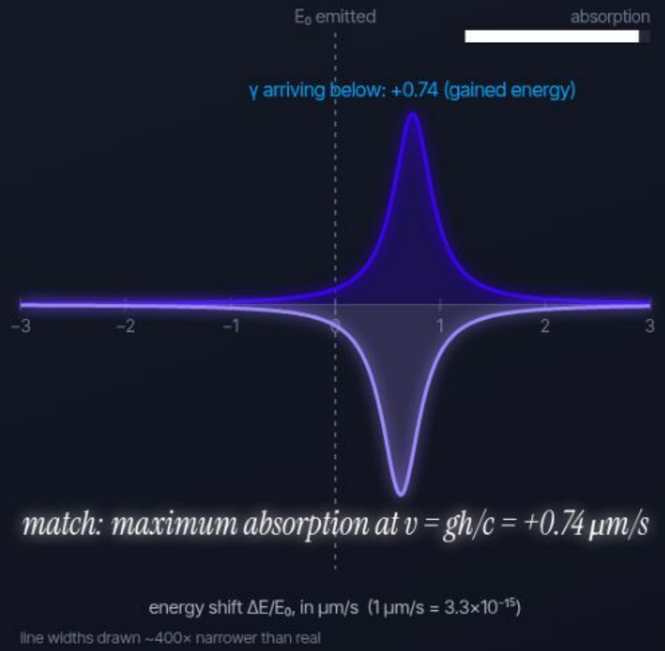
On the left, the star as it would look without gravity; on the right, ray-traced. The grid is the same; gravity bends the light around the star and reveals the **far hemisphere**.

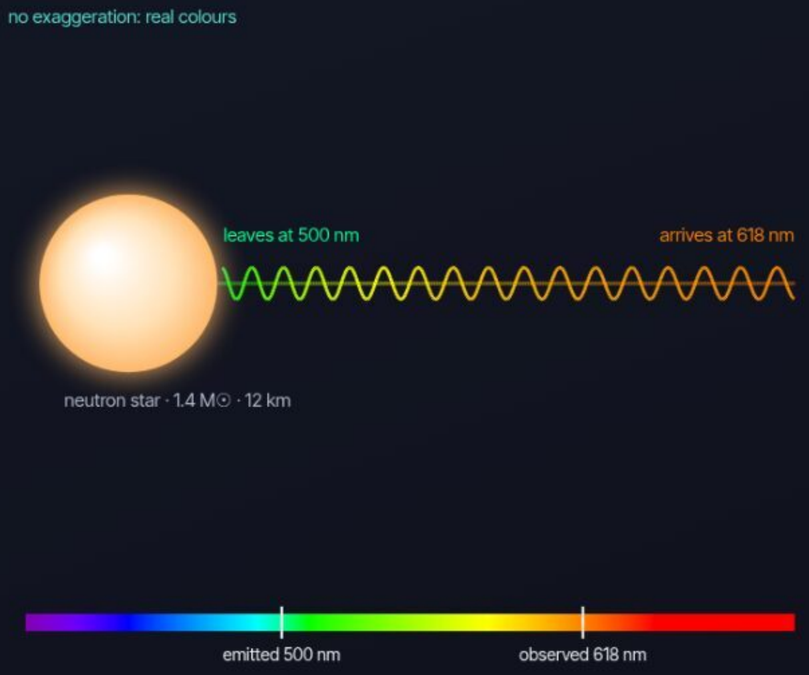
- The image is larger: apparent radius $R/\sqrt{(1 - R_S/R)}$.
- For $R < 1.76\ R_S$ the whole surface is visible — the far pole becomes a ring.
- Below $1.5\ R_S$ the star would lie inside the photon sphere.

1959 · POUND & REBKA · JEFFERSON TOWER, HARVARD

Light *loses energy as it climbs*







tower $\Delta\nu/\nu = gh/c^2$ `2.46×10⁻¹⁵` resonance at `+0,74 μm/s` ≈ 2.7 mm per hour absorber now `0.62 μm/s` star $1 + z = (1 - r_s/R)^{-½}$ `1.235`

14.4 keV γ rays from ^{57}Fe (Mössbauer effect). Pound & Rebka, PRL 4, 337 (1960): ~10%; Pound & Snider, Phys. Rev. 140, B788 (1965): ~1%. Every 12 s source and absorber swap places, as in the experiment.

The spectrum that *climbs out of the well*

Along a ray, I_ν/ν^3 is conserved. With $g = \sqrt{(1 - R_S/R)}$:

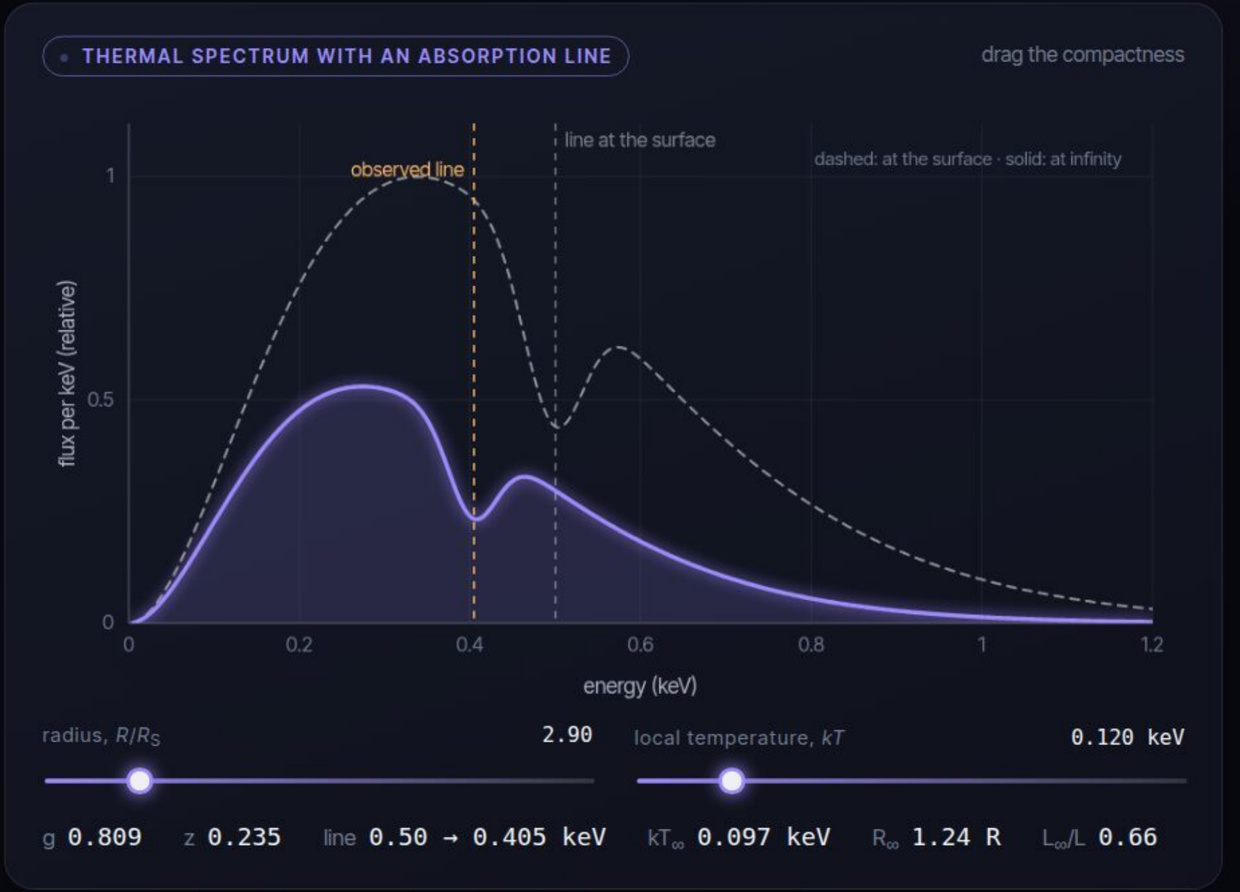
$$I_{obs}(E) = g^3 I_{em}(E/g)$$

SPECIFIC INTENSITY

$T_\infty = g\,T$ $\qquad$ $R_\infty = R/g$

A black body stays a black body, only cooler. The luminosity seen from afar is $L_\infty = g^2\,L$.

A **spectral line** of known energy E_0 arrives at $g\,E_0$: measuring the line means measuring the **compactness**.



PART I IN SUMMARY

The *ray tracing* recipe

1. **Choose the exterior metric**: Schwarzschild; Hartle–Thorne or Kerr with rotation; a different one for a different theory of gravity.
2. **Build the table** $\psi(b)$ and the delay $\Delta t(b)$ by integrating the null geodesics once.
3. **For each pixel** (or each surface element): $b \to$ emitting point, angle α, redshift g, delay.
4. **Intensity**: $I_{obs} = g^3\, I(E/g, \alpha')$, with the atmosphere's beaming and the Doppler shift from rotation.
5. **Add up**: image, flux — and, phase by phase, the **light curve** and the phase-resolved spectrum.

EXACT OR APPROXIMATE?

- **Exact integral** (or elliptic integrals): always valid, secondary images included.
- **Beloborodov (2002)** and Poutanen & Beloborodov (2006): closed formulae for ψ and for the delay, excellent for $R \gtrsim 3\ R_S$.
- **Oblate Schwarzschild** (Morsink et al. 2007): a star flattened by rotation, for millisecond pulsars.

This is the engine behind NICER's mass and radius measurements — and behind the results of Part III.

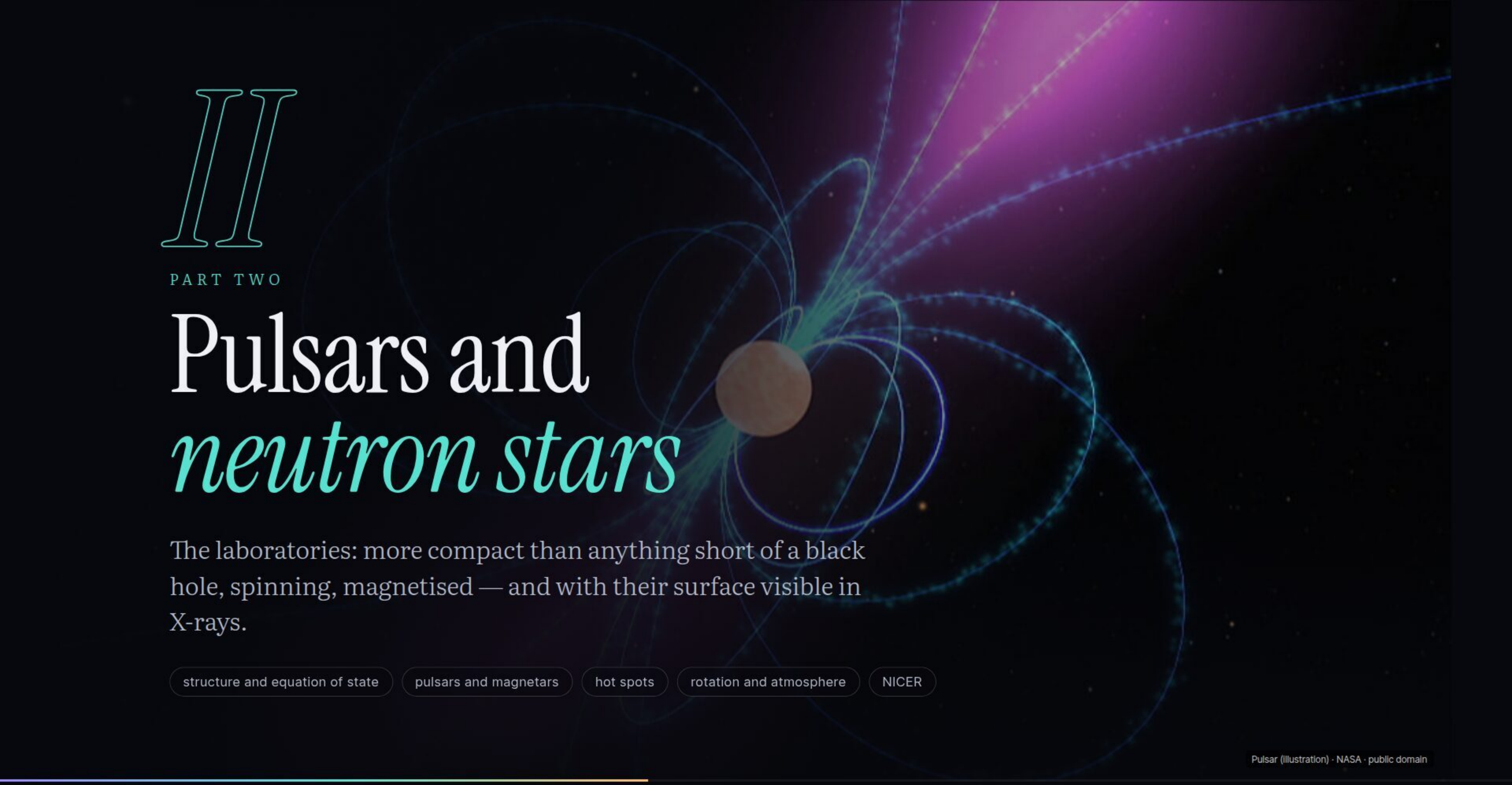
II

PART TWO

Pulsars and *neutron stars*

The laboratories: more compact than anything short of a black hole, spinning, magnetised — and with their surface visible in X-rays.

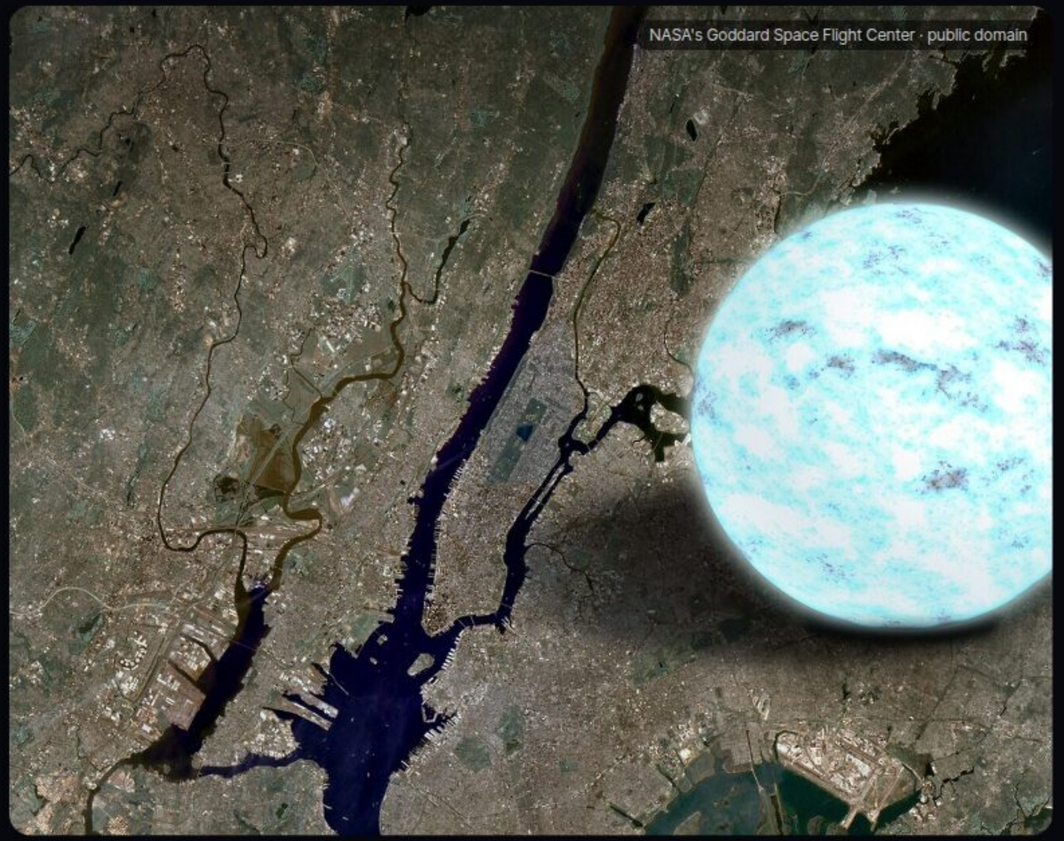
structure and equation of state | pulsars and magnetars | hot spots | rotation and atmosphere | NICER



Pulsar (illustration) · NASA · public domain

NEUTRON STARS IN NUMBERS

A solar mass *in a city*



A neutron star ~20 km across over Manhattan, to scale.

1.1–2.1 $M_\odot$
measured masses

11–13 km
radius (NICER, gravitational waves)

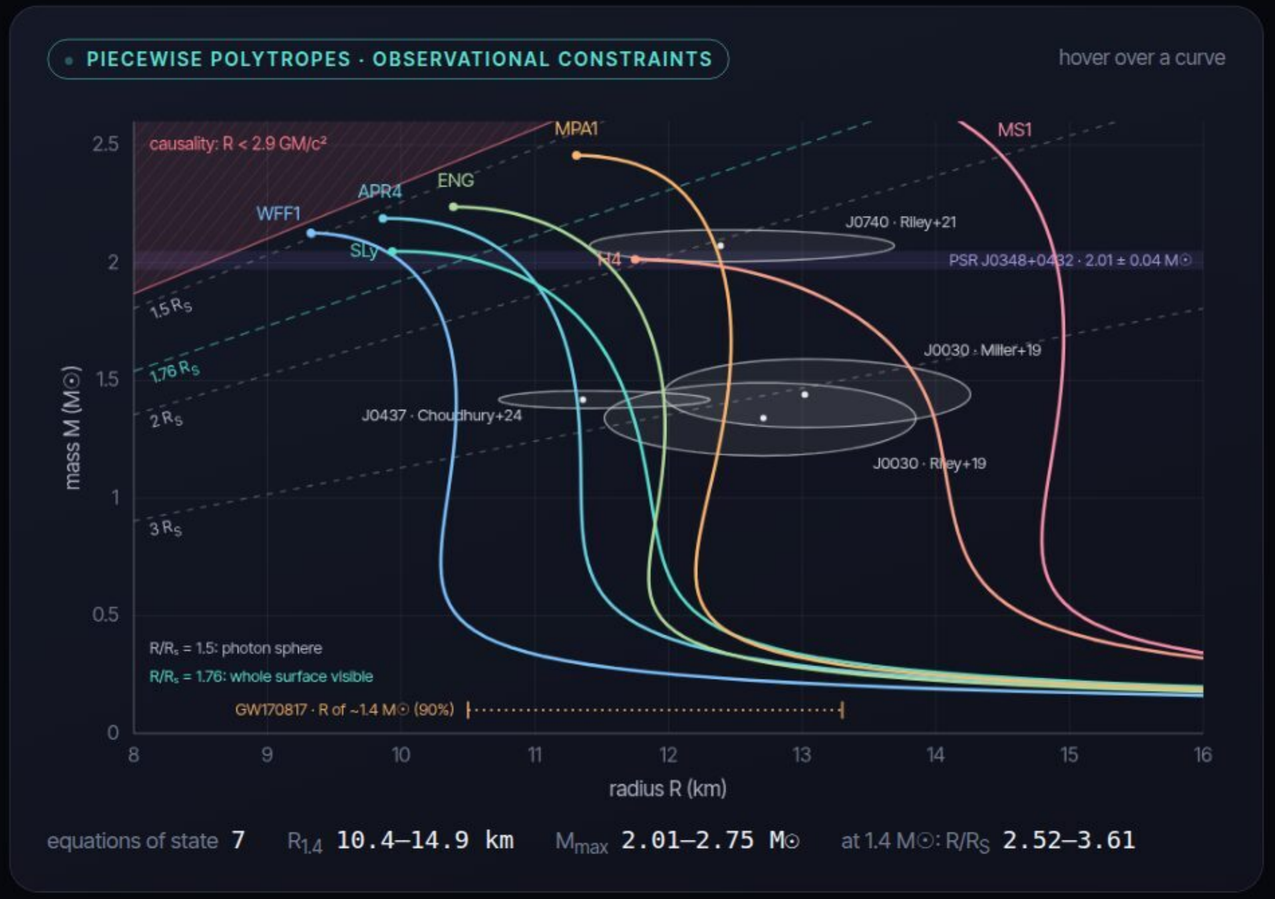
2–3 R_S
compactness: the regime of Part I

10^8–10^{15} G
surface magnetic field

Central density a few times that of the atomic nucleus, ~10^{15} g/cm^3. Spin periods from **1.4 ms** to tens of seconds.

Surface gravity is ~10^{11} times the Earth's; the escape velocity, 0.5 to 0.7 **c**.

From nuclear physics *to the M–R curve*



$$\frac{dP}{dr} = -\frac{G(\varepsilon + P)(m + 4\pi r^3 P/c^2)}{c^2 r(r - 2Gm/c^2)}$$

RELATIVISTIC HYDROSTATIC EQUILIBRIUM (TOV)

Each equation of state P(ε) gives a mass–radius curve and a maximum mass. Measuring the **mass and radius of the same star** picks between them.

The lines of constant compactness link the curve to Part I: at R = 1.76 R_S the whole surface would be visible; at 1.5 R_S, the photon sphere.

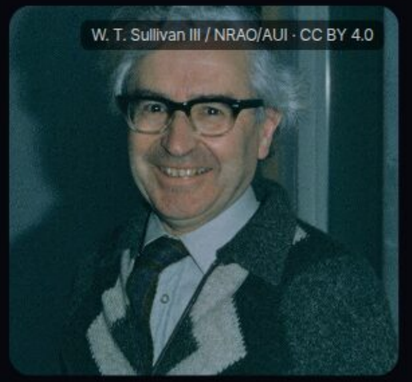
Piecewise polytropes of Read, Lackey, Owen & Friedman (2009); constraints (68%): NICER — Riley et al. 2019, 2021; Miller et al. 2019; Choudhury et al. 2024; PSR J0348+0432 (Antoniadis et al. 2013); GW170817 (Abbott et al. 2018).

CAMBRIDGE · 28 NOVEMBER 1967

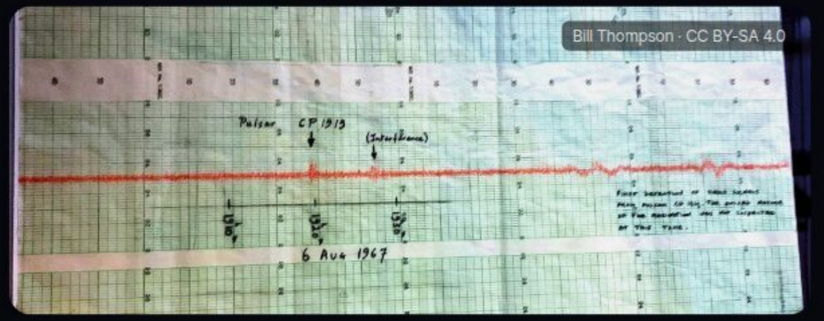
The first pulsar: *“LGM-1”*



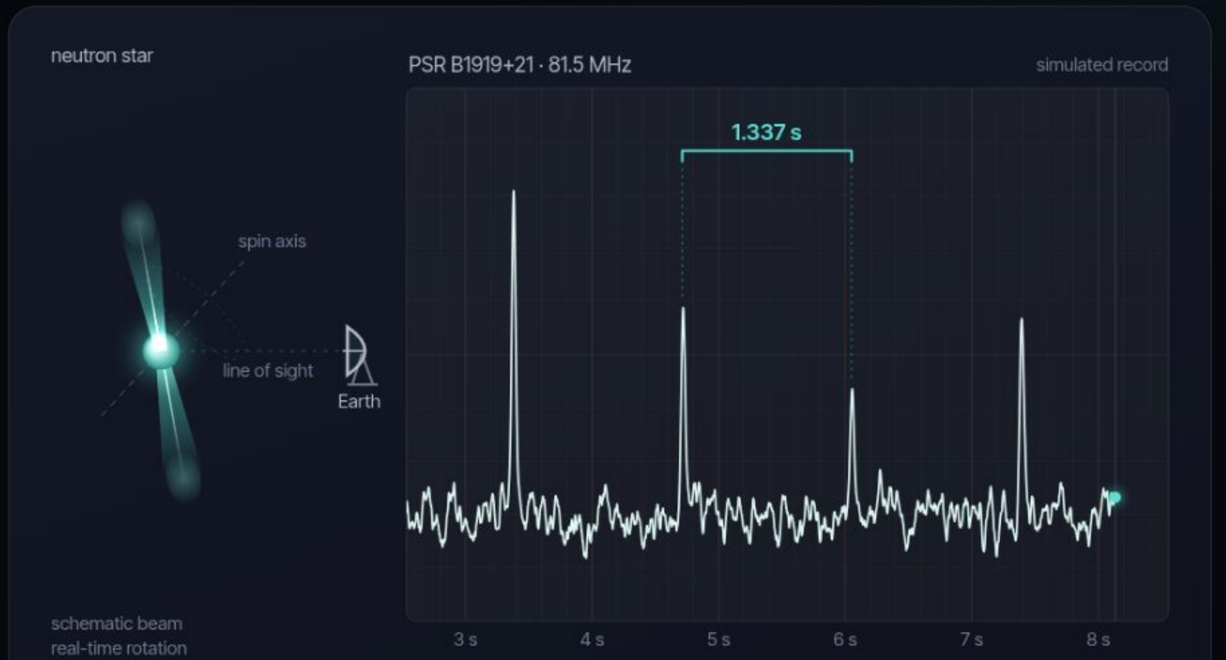
Jocelyn Bell
in 1967, PhD student



Antony Hewish
supervisor · Nobel Prize 1974



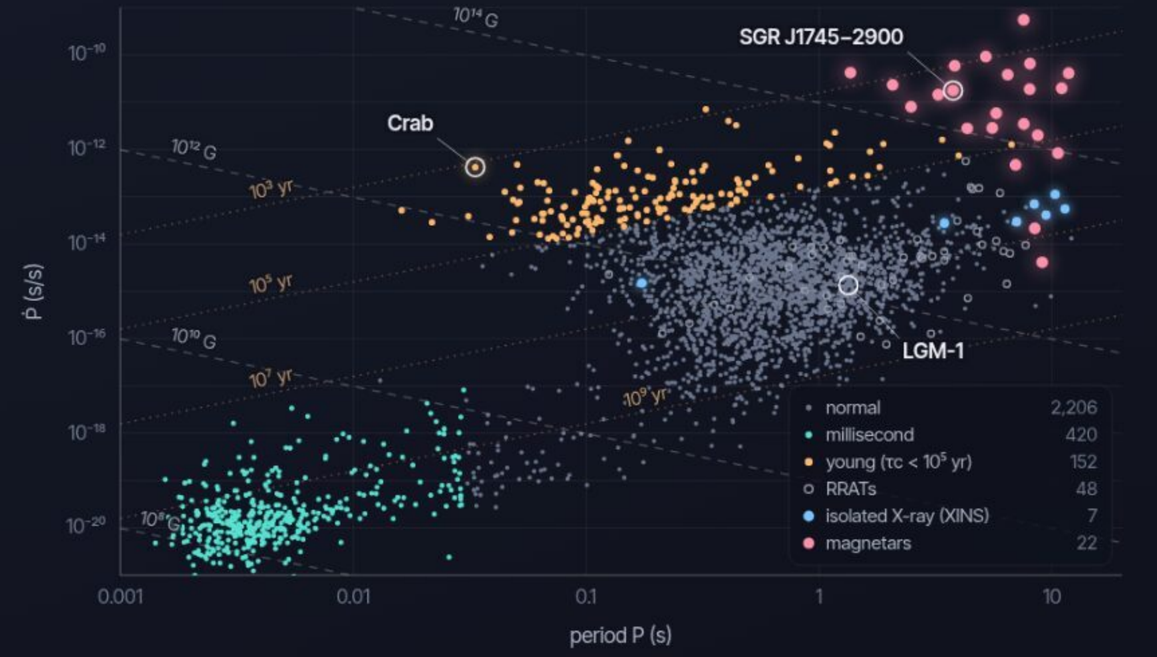
The original chart, August 1967: the “scruff” pencilled in as CP 1919.



period `1.337 302 s` $\dot{P}$ `1.35×10⁻¹⁵ s/s` frequency `81.5 MHz` pulses recorded `6`

Bell noticed a bit of “scruff” that kept coming back on the radio telescope’s paper chart: pulses every **1.337 s**, as regular as clockwork. Nickname: **LGM-1**, for *Little Green Men*. It was a spinning neutron star.

What is measured, *and what is inferred*



OBSERVABLE	EXPRESSION
period	P
characteristic age	$t_c = P/2\dot{P}$
magnetic field	$B = (3Ic^3P\dot{P}/8\pi^2R^6)^{1/2}$
rotational energy lost	$\dot{E}_{rot} = -4\pi^2I\dot{P}/P^3$

CANONICAL NEUTRON STAR

$M = 1.4\ M\odot \cdot R = 10\ km \cdot I = 10^{45}\ g\ cm^2$

$B = 3.2\times10^{19}\ (P\dot{P})^{1/2}\ G$

$|\dot{E}_{rot}| = 3.9\times10^{46}\ \dot{P}/P^3\ erg/s$

Only P and $\dot{P}$ are measured. The rest comes from assuming a spinning magnetic dipole — the straight lines in the diagram.

TRANSIENT ACTIVITY

A magnetar, *spinning*

SGR J1745−2900



field lines of a tilted dipole · real-time rotation

illustrative bursts

P 3.76 s B ≈ 2.6×10¹⁴ G characteristic age ≈ 3,400 yr $\dot{E}_{rot}$ ≈ 1.3×10³⁴ erg/s

bursts seen 1

EVENT	ENERGY
short bursts (~0.1 s)	10^{39} – 10^{41} erg
outbursts (months)	10^{41} – 10^{43} erg
giant flares	10^{44} – 10^{46} erg

The field lines, anchored in the crust, get twisted over time. When the crust can no longer take the stress, it cracks — a **starquake** — and the magnetic energy comes out all at once.

Giant flares in the Galaxy and the Magellanic Clouds: 1979, 1998 and 2004.

THE SURFACE IN X-RAYS

Why *hot spots* appear

MILLISECOND PULSARS

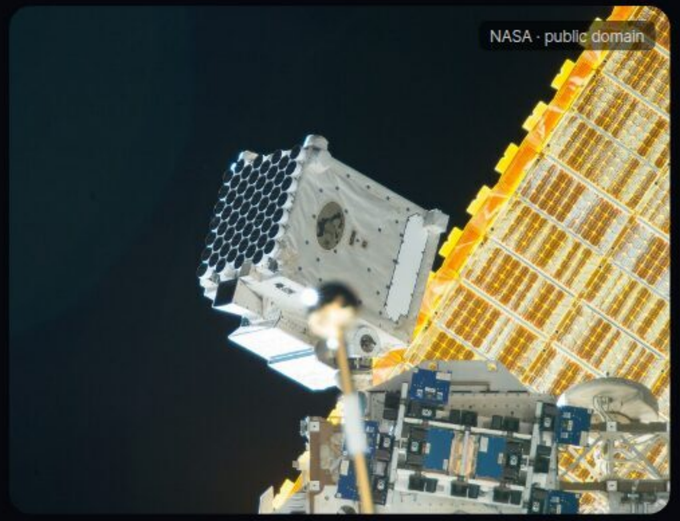
Particles accelerated in the magnetosphere flow back and **heat the polar caps** to ~10^6 K. They are NICER's targets.

MAGNETARS

Field decay and the currents along twisted field lines heat regions of the crust — which change after each *outburst*.

ISOLATED COOLING STARS

The magnetic field makes **heat conduction anisotropic**: poles hotter than the equator (the "Magnificent Seven").



NICER · on the Space Station since 2017 · ~100 ns time resolution



XMM-Newton · ESA, since 1999 · the largest area in soft X-rays



Chandra · NASA, since 1999 · the sharpest angular resolution

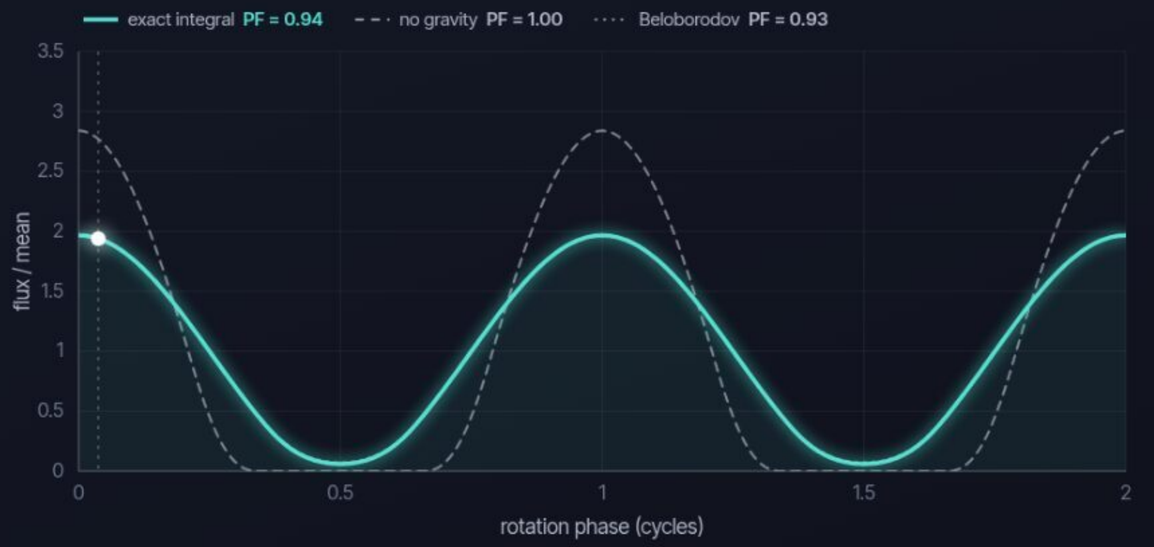
One spot, *one light curve*

STATIC STAR · EXACT INTEGRAL, BELOBORODOV OR FLAT SPACE

no gravity | Beloborodov | exact integral



static star · 1 turn per 4 s
sizes in units of R



radius, R/R_S 2.42

spot ↔ axis, i 60°

observer ↔ axis, j 75°

angular radius of the spot 20°

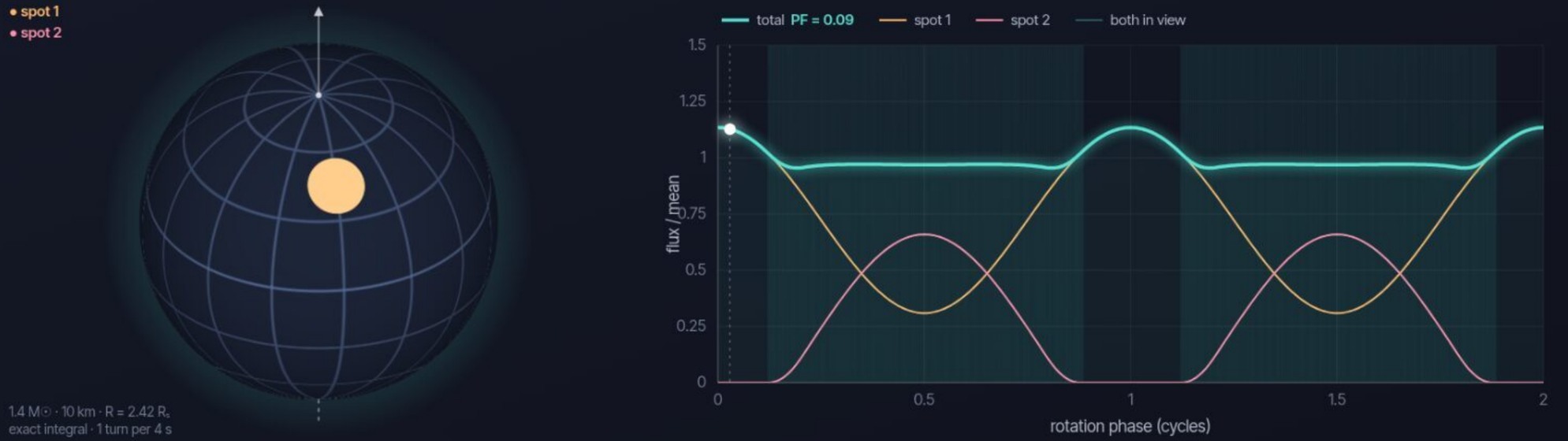
PF no gravity 1.000 PF Beloborodov 0.928 PF exact 0.939 surface visible 83% ψ_{max} 131.9° class B · centre hidden 10% of the turn (the rim stays)

secondary image no (needs R < 1.76 R_S)

Antipodal, *or not*

TWO SPOTS: ONE AT EACH MAGNETIC POLE

antipodal | offset



magnetic axis ↔ rotation, *i* 45°

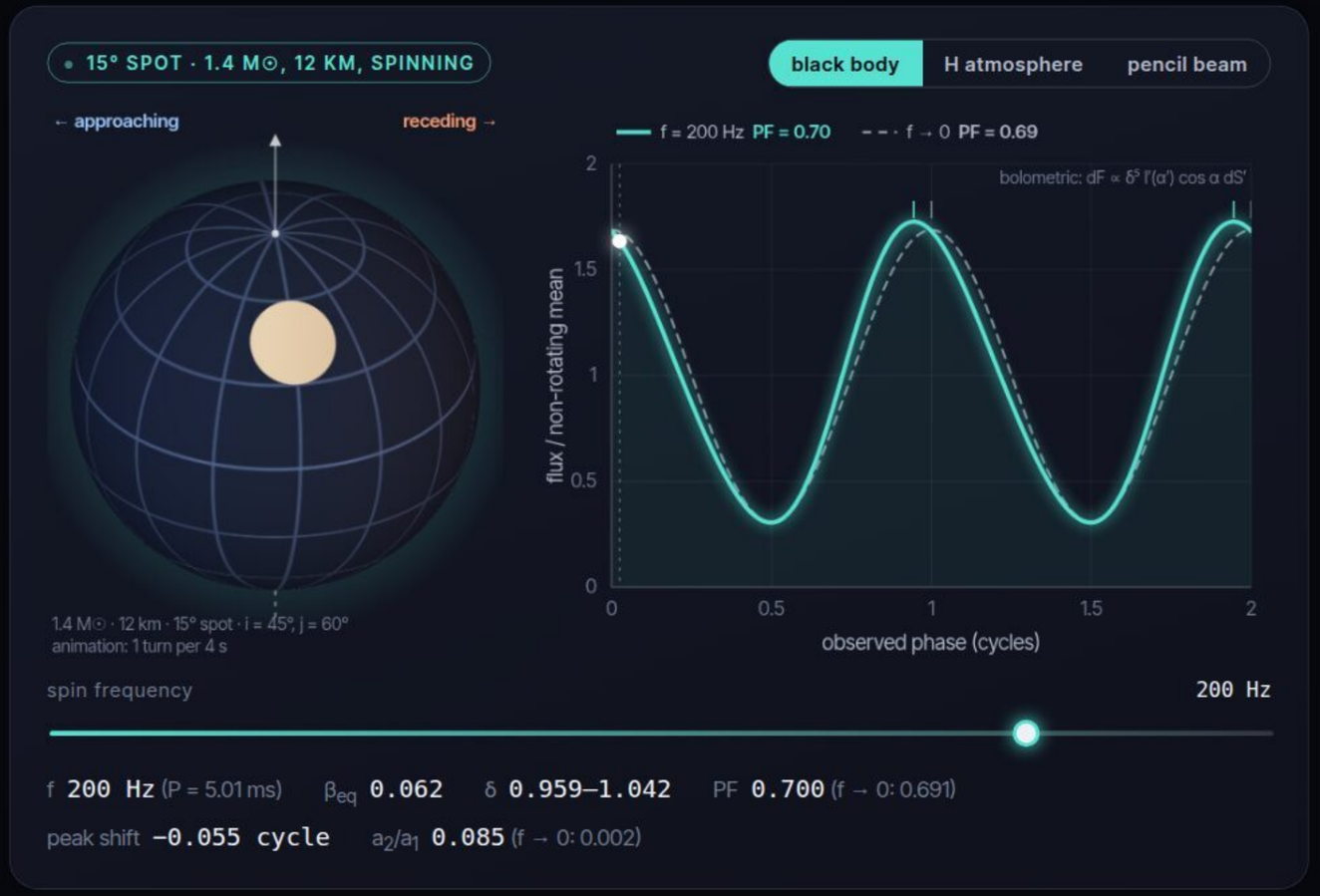
observer ↔ rotation, *j* 60°

offset of the second spot 0°

class II · primary always; antipodal sometimes PF 0.086 spot 1 in view always spot 2 in view 77% of the turn both 77% pulses separated by 0.50 cycle

Ψ_{max} 131.9°

Doppler, aberration, delays *and beaming*

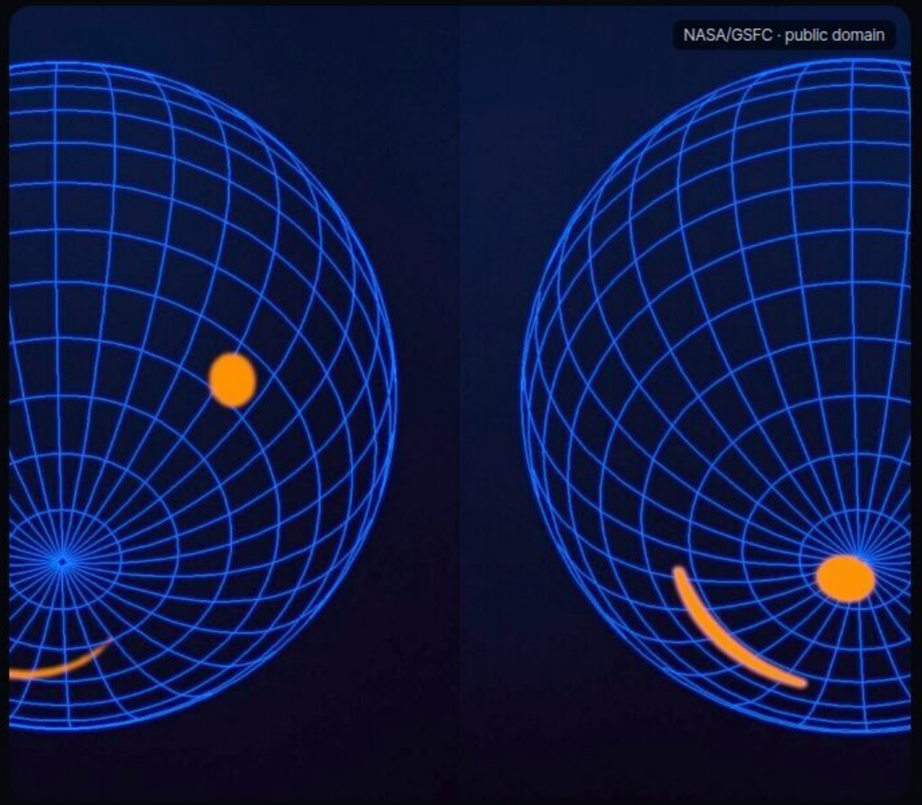


- **Doppler**: the surface moves at up to 0.1–0.2 c; the approaching side is brighter (bolometric flux ∝ δ^5).
- **Aberration**: the emission angle changes in the frame of the surface.
- **Delays**: tens of µs between points on the star — fractions of the period at hundreds of Hz.
- **Beaming**: an atmosphere does not emit equally in all directions; the profile depends on it.

Together they make the pulse **asymmetric** and create harmonics — and the asymmetry measures R separately from M.

NICER · PULSE-PROFILE MODELLING

Mass and radius *from the pulse shape*



PSR J0030+0451 · the solutions of the two teams: Amsterdam (Riley+ 2019, left) and Maryland (Miller+ 2019). In both, the spots are far from antipodal.

PULSAR	MASS ($M_\odot$)	RADIUS (KM)
PSR J0030+0451 · Riley+ 2019	$1.34^{+0.15}_{-0.16}$	$12.7^{+1.1}_{-1.2}$
PSR J0030+0451 · Miller+ 2019	$1.44^{+0.15}_{-0.14}$	$13.0^{+1.2}_{-1.1}$
PSR J0740+6620 · Riley+ 2021	$2.07^{+0.07}_{-0.07}$	$12.4^{+1.3}_{-1.0}$
PSR J0437−4715 · Choudhury+ 2024	1.42 ± 0.04	$11.4^{+1.0}_{-0.6}$

NICER time-stamps each photon to ~100 ns. Folding millions of photons at the period gives the pulse profile; ray tracing, with M, R, the geometry and the spots left free, fits that profile by Bayesian inference.

J0030's spots came out non-antipodal and elongated: the field is not a centred dipole.

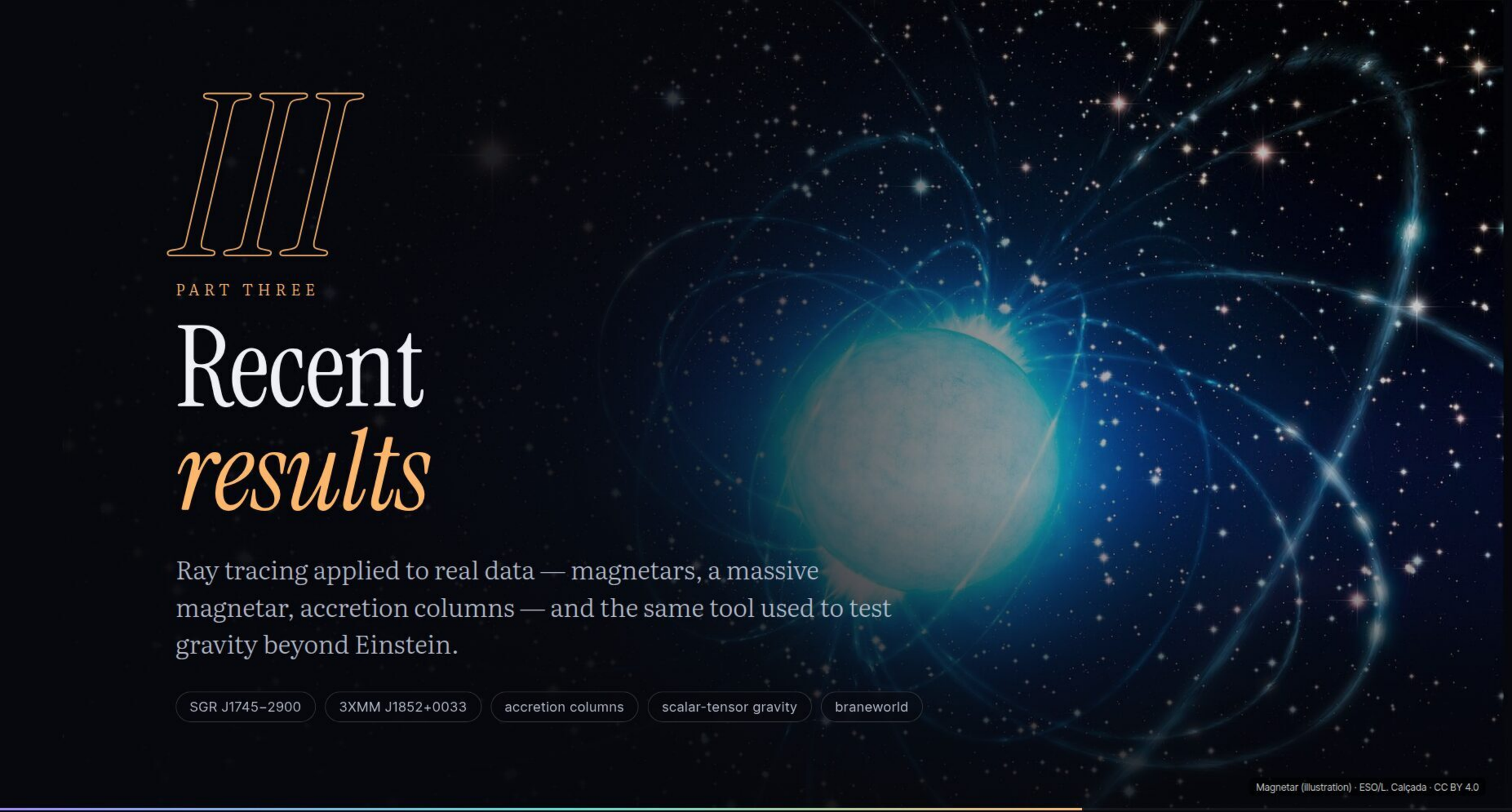
III

PART THREE

Recent *results*

Ray tracing applied to real data — magnetars, a massive magnetar, accretion columns — and the same tool used to test gravity beyond Einstein.

SGR J1745−2900 · 3XMM J1852+0033 · accretion columns · scalar-tensor gravity · braneworld



Magnetar (illustration) · ESO/L. Calçada · CC BY 4.0

2017 – 2026

Light curves *as an instrument*

2017

A&A 599, A87: rotation-powered SGRs and AXPs, with realistic neutron stars

2020

ApJ 889, 165: SGR J1745–2900 and a multipolar field, from the light curve

2023

Galaxies 11, 18: pulse profiles of accretion columns

2024

JHEAp 42, 52: 3XMM J1852+0033, a massive low-field magnetar

2024

EPJC 84, 1337: light curves as a test of scalar-tensor gravity

2024

PRD 110, 103014: stratified rotation and the moment of inertia

2025

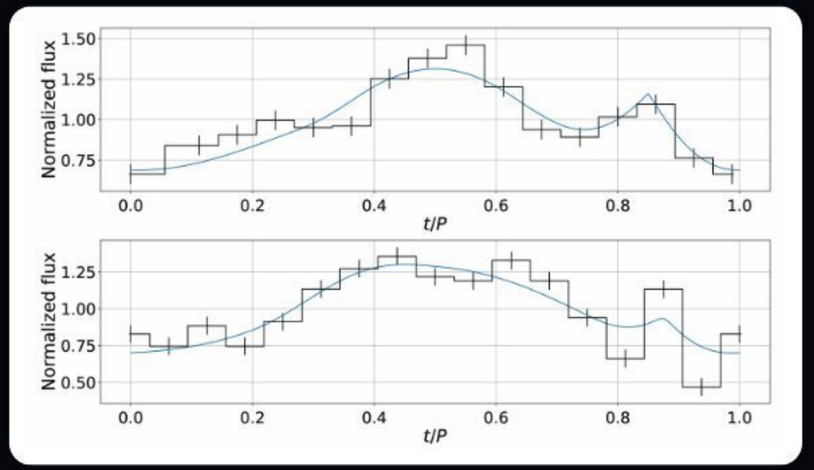
PRD 112, 083010: quark stars and the QCD equation of state

2026

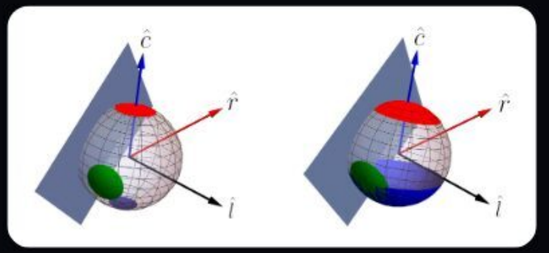
arXiv:2607.09836: compact stars in a high-tension braneworld

The common thread: reading in the light of a compact star its *mass*, its *radius*, its *field* — and the *gravity* that curves it.

SGR J1745−2900: *more than two spots*



Chandra light curves (2013 and 2016) and the three-spot fits.

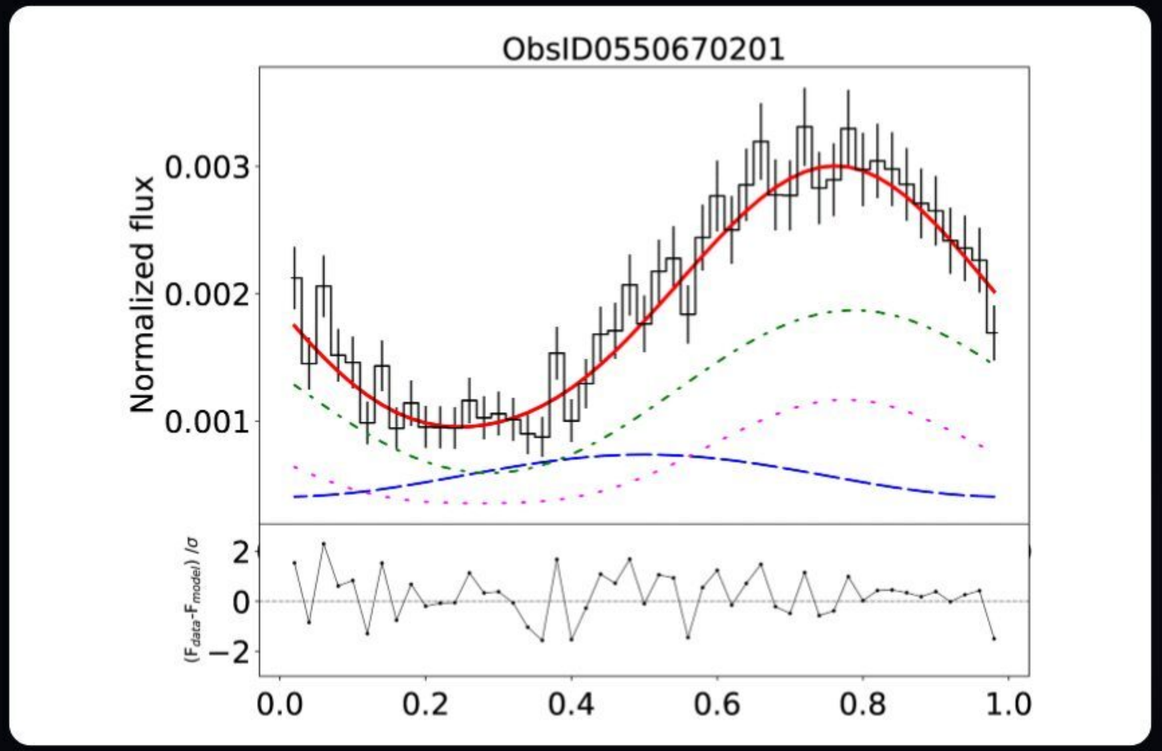


The three spots: two at the poles, one in the southern hemisphere.

- The magnetar 0.1 pc from Sgr A*; 3.5 years of **Chandra** data.
- Model: M, R, up to **three spots** of any size and position, Schwarzschild light bending; fitted with a genetic algorithm.
- Two or three spots describe the data; with three, they are **large** (16–67°).
- A field with non-trivial poloidal and toroidal structure — the same message NICER gave for J0030.

1.4 ± 0.1	10.9 ± 1.5
$M_\odot$	km

3XMM J1852+0033: *ray tracing and inference*



One of the five pulse profiles (XMM-Newton, 2008) and the best fit: dashed, the contribution of each spot.

- A **transient magnetar** near the remnant Kes 79, P = 11.56 s; until then there were only upper limits on the field.
- Five **XMM-Newton** epochs over three weeks in 2008.
- **Ray tracing** in Schwarzschild with circular spots of any size (Turolla & Nobili 2013) and **magnetised atmospheres** (NSMAX: H, C, O), with anisotropic emission.
- **Bayesian inference** (MCMC): mass, radius, field and spot positions shared by all epochs; size and temperature free to vary.
- Tested by **injection and recovery** of simulated data.

BEST MODEL

Three small spots, two in the southern hemisphere, and a **carbon** atmosphere: $\chi^2/\mathrm{dof} = 1.22$, well ahead of two spots or of hydrogen.

A *massive*, low-field magnetar

$2.09^{+0.16}_{-0.09}\ M_\odot$

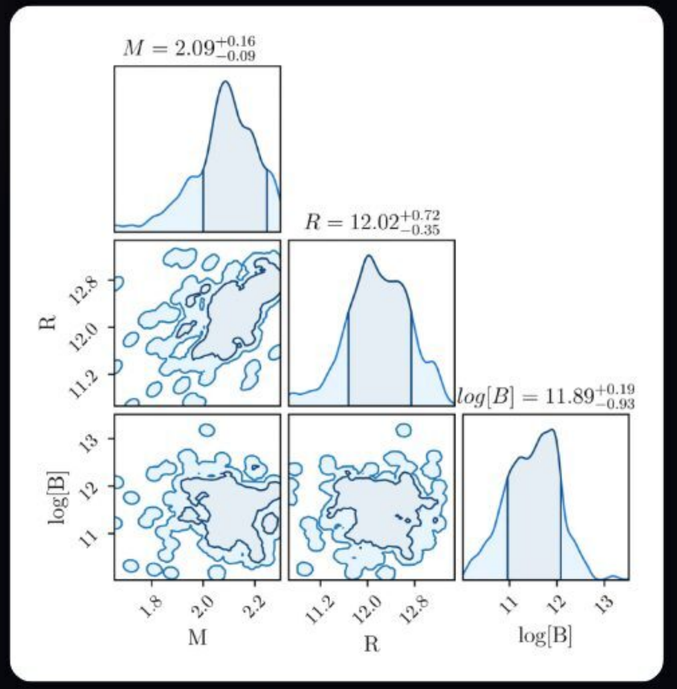
mass (1σ): among the highest ever estimated

12.0 ± 1.4 km

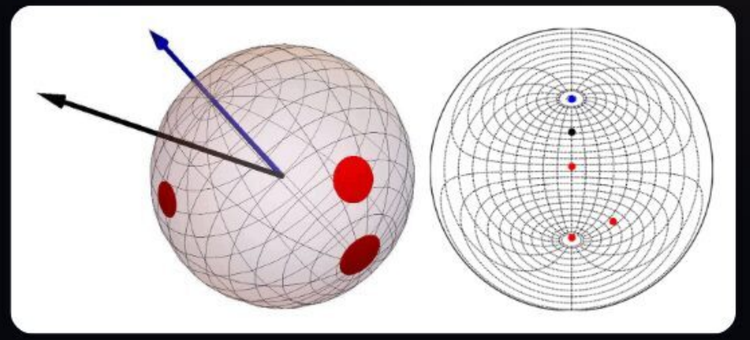
radius (2σ)

$\sim 8 \times 10^{11}$ G

surface field: log B = $11.89^{+0.19}_{-0.93}$, below the critical value



Posteriors of *M*, *R* and log *B*.

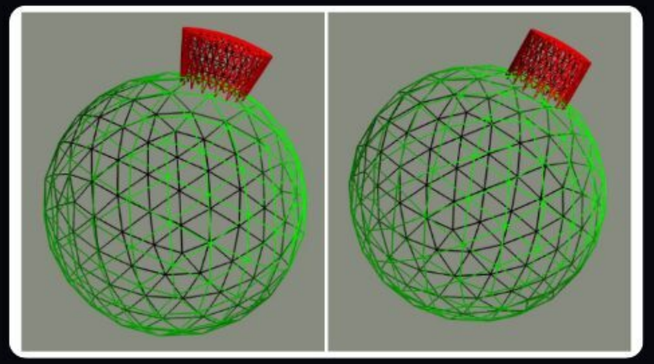


The best-fit geometry.

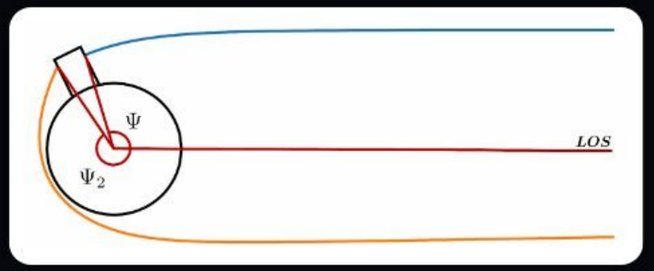
Interpretation: accretion of supernova material **buried** the field, which is now re-emerging; the carbon atmosphere would come from diffusive nuclear burning of H/He.

Compactness $GM/Rc^2 \approx 0.26$: favours equations of state of intermediate stiffness.

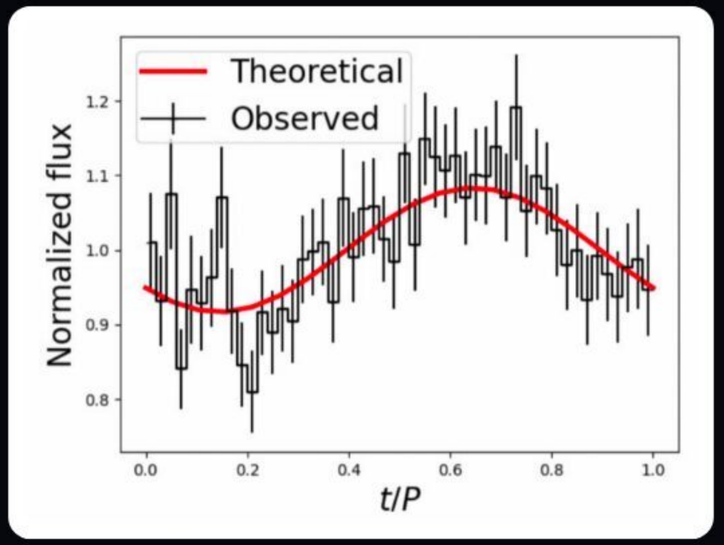
When the light comes *from a column*



The star and the columns (truncated cone and cylinder) as a 3D mesh.



Photons emitted above the surface may pass through a **periastron** before escaping.



1A 0535+262 (NuSTAR, 2018): observed profile and the two-column model.

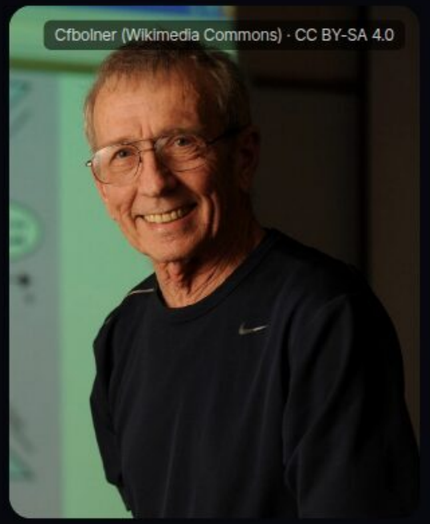
- In X-ray binaries, matter falls along the field lines and forms **columns** above the poles: the emission comes out sideways (fan beam).
- **Exact** ray tracing in Schwarzschild, tabulated, including for points above the surface; Doppler included.
- Genetic algorithm for the priors, then MCMC.

$1.53^{+0.52}_{-0.40}$ $M_\odot$

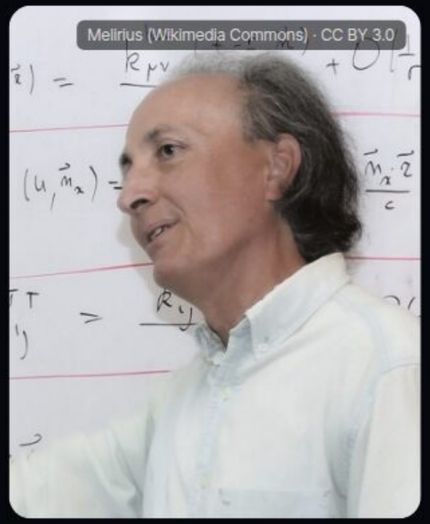
$12.1^{+1.1}_{-0.9}$ km

BRANS & DICKE 1961 · DAMOUR & ESPOSITO-FARÈSE 1993

What if gravity *is not Einstein's?*



Carl Brans
1935 – 2022



Thibault Damour
b. 1951

In **scalar-tensor** theories, besides the metric there is a scalar field ϕ that couples to matter. In the Solar System the effect is tiny; in a neutron star **spontaneous scalarisation** can occur: the field grows on its own.

Outside the star, the metric is no longer Schwarzschild: it is the **Just metric**, with a scalar charge Q:

$$ds^2 = -(1 - \frac{a}{r})^{b/a} dt^2 + (1 - \frac{a}{r})^{-b/a} dr^2 + (1 - \frac{a}{r})^{1-b/a} r^2 d\Omega^2$$

$B = 2GM/C^2 \cdot A/B = \sqrt{(1 + Q^2)}$ · EINSTEIN FRAME

Two stars with the **same mass and the same radius** — "doppelgängers" — bend light in different ways. The light curve can tell the theories apart.

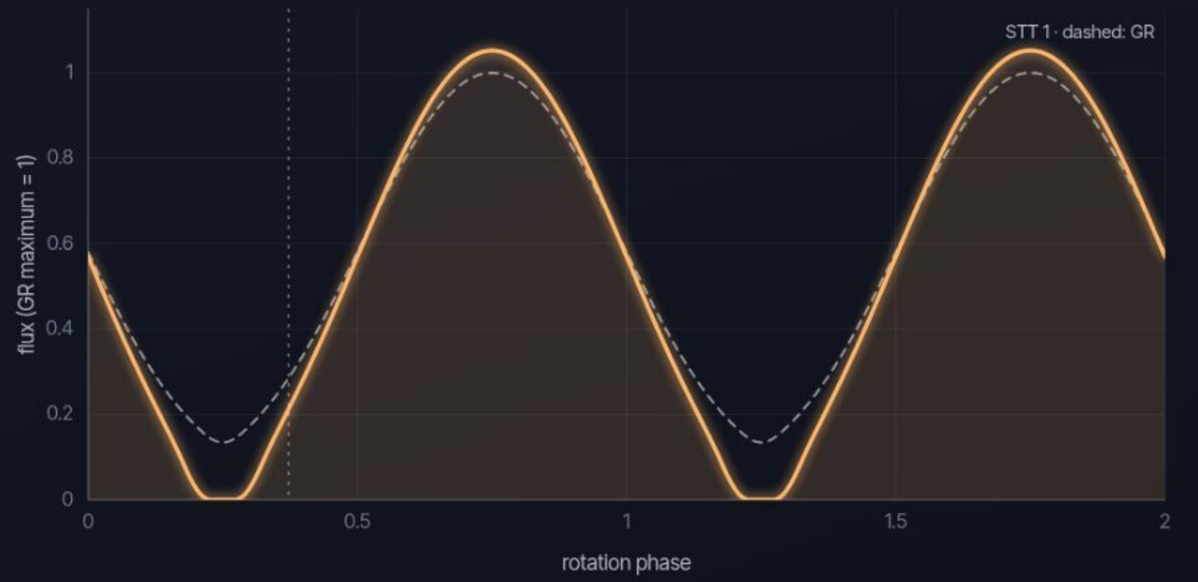
The same star, *different gravity*

• M = 2.1 M⊙ · ONE 10° SPOT · I = Θ = 90° · STATIC STAR · EXACT INTEGRAL

GR | STT 1 · A_s = 0.95 | STT 2 · A_s = 1.05

the star seen from afar · dashed: its size in GR



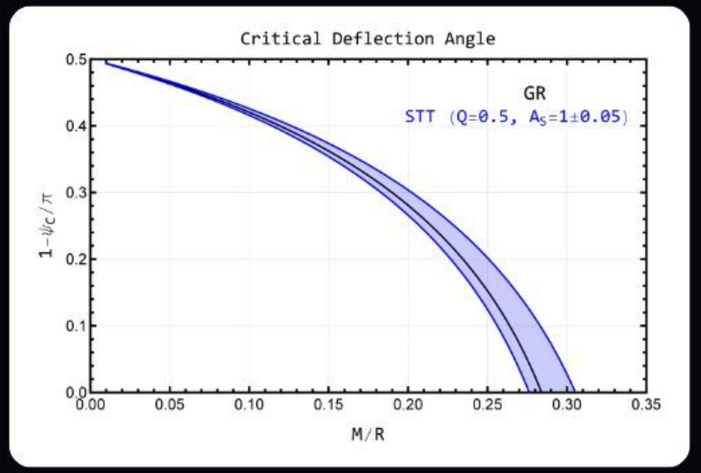


compactness, GM/Rc^2 0.284

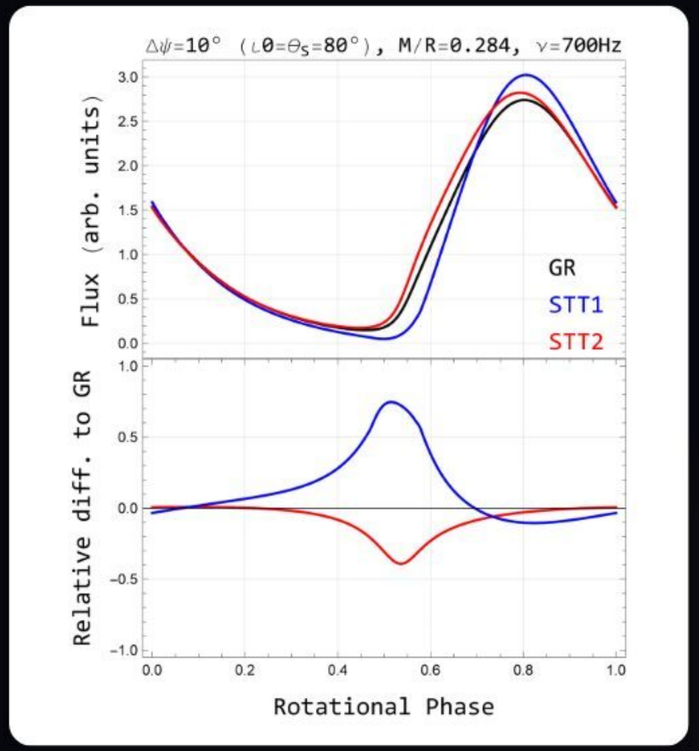
scalar charge, Q 0.50

STT 1: ψ_c 0.903π GR: ψ_c 1.000π invisible zone 17.4° R 10.92 km apparent radius 16.88 km PF 1.00

Up to *80%* difference in flux



The critical angle versus *M*/*R*: the GR line and the scalar-tensor band.



Spinning at 700 Hz, with Doppler and delays: flux and difference relative to GR.

- The compactness at which $\psi_c = \pi$ ranges from **0.275 to 0.305** in scalar-tensor theory, against 0.284 in GR.
- The flux from a spot differs from GR by up to **~80%**.
- The radius changes by up to **~10%** for the same stellar parameters: today, this limits $Q \lesssim 0.5$.
- With NICER, J0740+6620 has $C = 0.21$–0.28: not yet enough to separate the theories.

OUTLOOK

Radii with smaller uncertainties — eXTP, NewAthena, and gravitational waves from Einstein Telescope and LISA — may constrain scalar-tensor theories through the **pulse shape**.

Compact stars *in a braneworld*

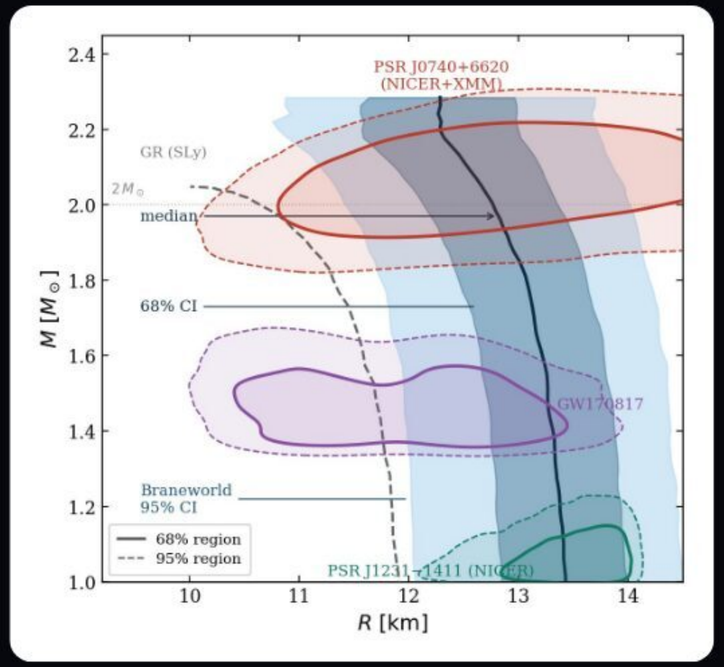
In the braneworld, our Universe is a membrane in an extra dimension. In the star, this shows up as corrections to the TOV equations:

$$\rho_{\rm eff} = \rho + \frac{\rho^2}{2\lambda} + \alpha_{\mathscr{W}}\rho$$

$$p_{\rm eff} = p + \frac{p\rho}{\lambda} + \frac{\rho^2}{2\lambda} + w_{\mathscr{W}}\alpha_{\mathscr{W}}\rho$$

Λ: BRANE TENSION · $A_{\mathscr{W}}$, $W_{\mathscr{W}}$: THE WEYL SECTOR

MCMC with the **NICER** radii (J0740+6620, J1231−1411 — themselves ray-tracing results) and GW170817, for the SLy equation of state.



The braneworld M–R band (68% and 95%), GR (dashed) and the NICER and GW170817 constraints.

$-0.15^{+0.30}_{-0.08}$

$\alpha_{\mathscr{W}}$: slightly negative Weyl coupling

$10^{4.0 \pm 1.5}$

λ (km^{-2}): high tension

$2.30^{+0.14}_{-0.08}$

$M_{\rm max}$ ($M\odot$) · GR+SLy: 2.05

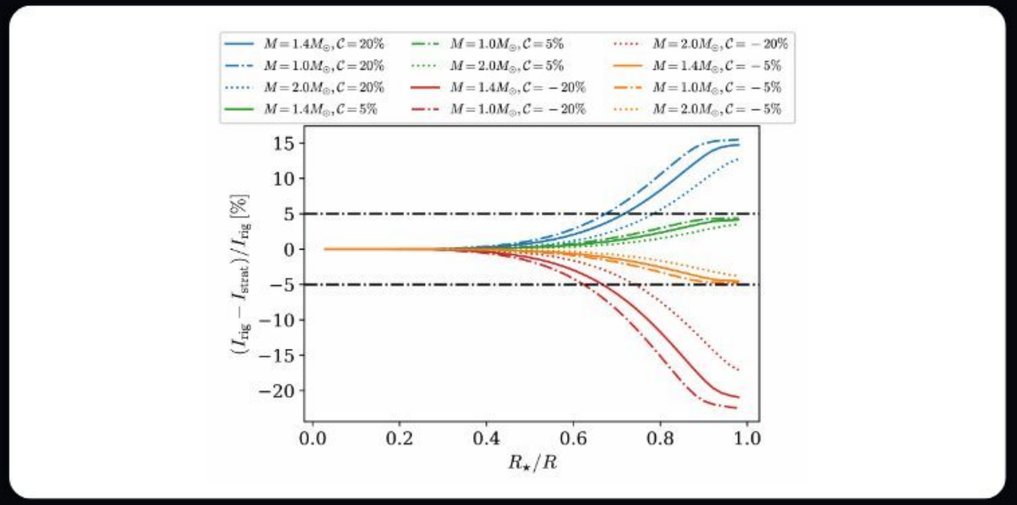
13.3 ± 0.6

$R_{1.4}$ (km) · GR+SLy: 11.8

Consistent with the data, with no large departures from GR. Preprint, under review.

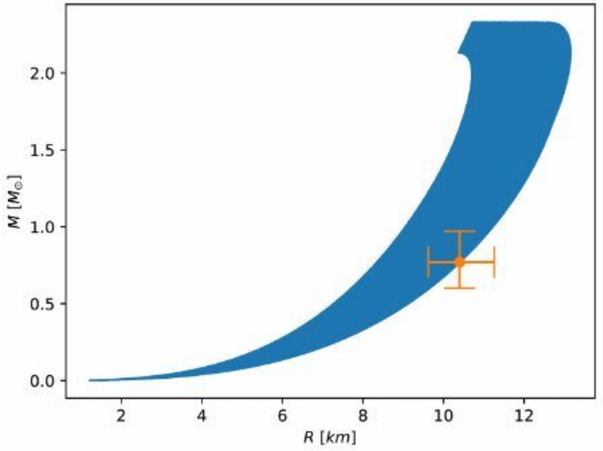
PRD 110, 103014 (2024) · PRD 112, 083010 (2025)

What goes into *M, R and I*



ΔI/I against the position *R*/R* of a jump in angular velocity.

Stratified rotation (Pereira, Ottoni, Coelho, Rueda & de Lima): if the interior rotates differently from the crust, the moment of inertia changes — little if the jump is deep (< 40% of the radius), almost as much as Δω if it is near the surface. This affects the rotational energy budget of pulsars and weakens the universal I-Love-Q relations.



The allowed M–R band for quark stars, with XMMU J173203.3−344518.

Quark stars (Zastrow, Pereira, de Lima & Horvath): a QCD-inspired equation of state, scanned against maximum masses, radii and tidal deformability. The strong interactions must deviate by ≥ 20% from free quarks; a 1.4 M⊙ star would be 10.0–12.3 km in radius.

CONCLUSIONS

From geometry *to observation* — and back

- In a neutron star, light bending, redshift and delays are **not corrections**: they define the image and the light curve.
- Backward ray tracing — with the exact integral when the compactness calls for it — links the metric to what a telescope measures.
- Applied to Chandra, XMM-Newton and NuSTAR data, it gave the mass, radius, field and spot geometry of magnetars and of an accreting star.
- Since the light curve depends only on the exterior metric, it also **tests gravity**: scalar-tensor theories and braneworld.

NEXT STEPS

- Magnetised atmospheres and exact ray tracing together, for isolated cooling stars.
- Fast rotation and metrics beyond Schwarzschild in the same fits.
- Data from the next generation: eXTP, NewAthena, STROBE-X.

The panels in this seminar run in the browser: *rafael-lima.pages.dev/talks*.

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P–Ṗ diagram — ATNF pulsar catalogue 2.8.1 (Manchester et al., AJ 129, 1993, 2005)

THANK YOU

Questions?

Slides, animations and dashboards available at
rafael-lima.pages.dev/talks

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